# Hardware Bill of Materials (BOM) & Billed Budget

Production physical component pricing, GPIO hardware mappings, and subsystem budget roll-up.

| TOTAL BILLED COST | CORE MCU | SUBSYSTEMS | RIG STATUS |
|---|---|---|---|
| ■ 2,920 | ESP32 Dual-Core | 8 Subsystems | Verified & Billed |
| Full Functional Rig | 240MHz / 4MB Flash | Sensors, Power, Fluidics | Active Cultivation Ready |

## 1. Actual Billed Hardware Cost Breakdown

| # | Subsystem / Category | Included Components & Modules | Billed Cost (INR) |
|---|---|---|---|
| 1 | **Compute & Controller** | ESP32-WROOM-32 Dev Board (■300) + Terminal Expansion Board (■120) | **■ 420** |
| 2 | **Biophysical Sensors** | TCS34725 RGB Colorimeter, BME280 Ambient, DS18B20 Temp, JSN-SR04T Sonar, DS3231 RTC | **■ 300** |
| 3 | **Displays & HMI** | 1.3" I2C OLED Display (SH1106/SSD1306) + Tactile Push Buttons (NAV / ACTION) | **■ 300** |
| 4 | **Relays, SSR & Switching** | 4-Ch Relay Board (Songle), 1-Ch Relay Board, Solid State Relay (SSR for aeration) | **■ 400** |
| 5 | **Actuators & Fluidics** | 12V Intake Solenoid Valve, Outtake Drain Pump, Aeration Sparger Pump, Agitator Motor, Grow LEDs, Cooling Fan | **■ 300** |
| 6 | **Power Supply & Regulation** | 12V DC SMPS / Power Supply Adapter + LM2596 DC-DC Step-Down Buck Converter (5V 3A) | **■ 300** |
| 7 | **Passives & Interconnects** | 1N4007 Flyback Diodes, Pull-up Resistors (4.7kΩ), Voltage Divider, Terminal Blocks, Jumper Wires | **■ 100** |
| 8 | **Vessel, Optics & Enclosure** | Cultivation Column Tank (33cm depth), Optical Light-Tight Shroud, Silicone Tubing, Micro Diffuser, IP65 Project Box | **■ 800** |
| | **TOTAL BILLED HARDWARE BUDGET** | **Complete 8-Subsystem Physical Bioreactor Rig** | **■ 2,920** |

**Cost Optimization Highlight:** By consolidating multiple laboratory instruments (spectrophotometer, timer relays, PID chiller loops) into an integrated ESP32 FreeRTOS controller with low-cost solid-state and optical components, total rig BOM cost is optimized to ■**2,920** while retaining full cloud IoT telemetry and autonomous safety interlocks.

## 2. Controller GPIO & Peripheral Matrix

| GPIO Pin | Peripheral / Subsystem | Interface | Operating Volts | Hardware Logic / State | Circuit Protection / Notes |
|---|---|---|---|---|---|
| **GPIO 21** | I2C Data (`SDA`) | I2C Shared | 3.3V | Open-drain | Shared bus: TCS34725, BME280, RTC, OLED |
| **GPIO 22** | I2C Clock (`SCL`) | I2C Shared | 3.3V | Open-drain | Hardware pull-up on bus |
| **GPIO 4** | Submersible Water Temp (DS18B20) | 1-Wire | 3.3V / 5V | Open-drain | **4.7kΩ pull-up resistor** to 3.3V required |
| **GPIO 18** | Ultrasonic Level `TRIG` | Digital Out | 3.3V / 5V | Active-HIGH | 10µs pulse to JSN-SR04T driver |
| **GPIO 19** | Ultrasonic Level `ECHO` | Digital In | 3.3V Safe | Active-HIGH | **Resistor divider** (1kΩ / 2kΩ) to protect ESP32 |
| **GPIO 33** | Media Intake Solenoid Valve | Relay 1 | 12V Load | Active-LOW | Flyback diode 1N4007 across coil |
| **GPIO 32** | Harvest Outtake Drain Pump | Relay 3 | 12V Load | Active-HIGH | Single-channel relay module; flyback diode |
| **GPIO 25** | Aeration Air Sparger Pump | SSR 1 | 12V / 230V | Active-LOW | Solid state relay for silent 5m/25m cycling |
| **GPIO 26** | Mechanical Agitator Motor | Relay 4 | 12V Load | Active-LOW | Drives wall-wiper every 4h; flyback diode |
| **GPIO 27** | Photoperiod Grow Light Array | Relay 2 | 12V Load | Active-LOW | Full-spectrum LED illumination (8:00–20:00) |
| **GPIO 23** | Chamber Exhaust Cooling Fan | Relay 5 | 12V Load | Active-LOW | Automatic heat evacuation (>34°C) |
| **GPIO 13** | OLED Navigation Button (NAV) | Digital In | 3.3V | Active-LOW | Internal INPUT_PULLUP; cycles display pages |
| **GPIO 14** | OLED Action Button (ACTION) | Digital In | 3.3V | Active-LOW | Internal INPUT_PULLUP; jog/dev mode toggle |

## 3. Critical Assembly & Safety Guidelines

**1. Inductive Flyback Suppression:** Install 1N4007 diodes across solenoid valve and DC motor terminals (cathode to +12V, anode to GND) to eliminate reverse-EMF transients that cause ESP32 watchdog reboots.

**2. 5V Ultrasonic Echo Clamping:** The JSN-SR04T receiver outputs a 5V pulse on ECHO. The 1kΩ / 2kΩ resistive divider ensures the ESP32 GPIO 19 never sees voltages exceeding 3.33V.

**3. Optical Biomass Isolation:** The TCS34725 color sensor must be seated flush against the vessel in an opaque 3D-printed shroud to ensure repeatable Green Excess Index (GEI) calculations unaffected by room lighting.

**4. Galvanic Separation:** Ensure high-current 12V pump/LED grounds and sensitive 3.3V sensor grounds meet only at a single star-ground point on the power terminal bus to prevent ground loops.